

Embedding-based prediction of brain responses reveals emotional effects in sentence processing

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Introduction

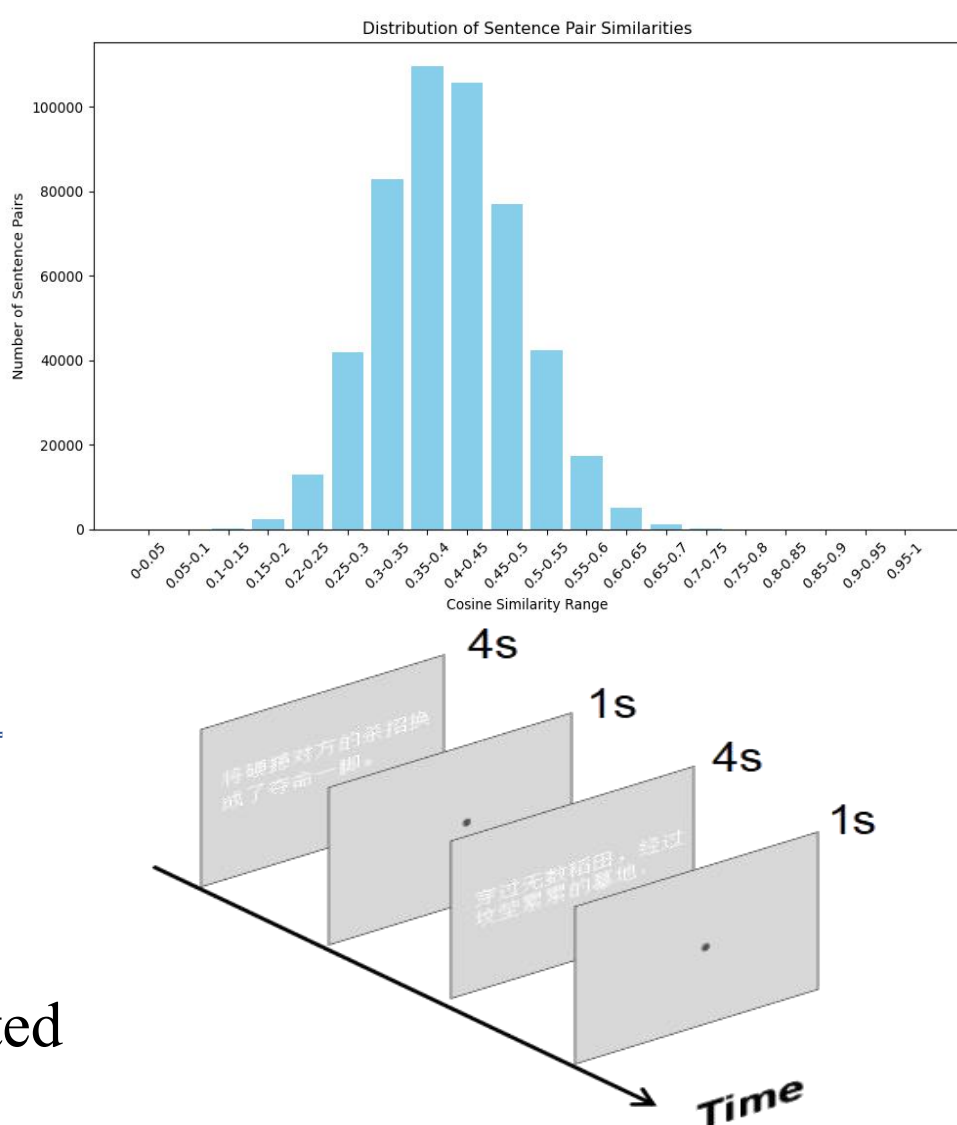
- Reading different sentences evokes different feelings and thoughts
- Large language models (LLMs) share many aspects of language processing with the human brain and can be used to **predict neural responses** to linguistic inputs

Present Question

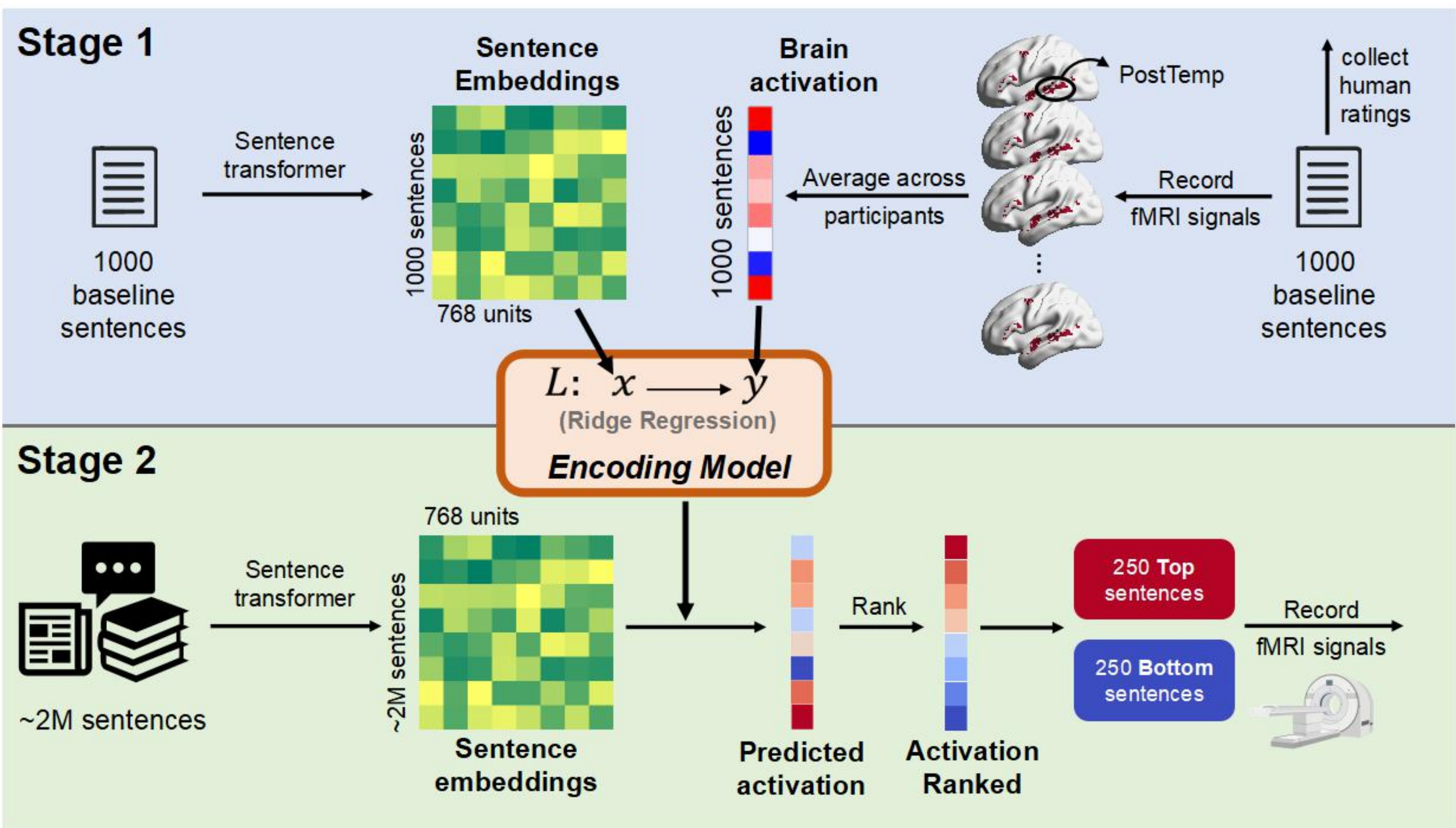
1. Can LLMs **predict and modulate** neural responses to linguistic inputs?
2. What **specific semantic features or dimensions of sentence** meaning are employed by both humans and LLMs to encode semantic information?

Design

- **Stage 1:** a 1000 sentences corpus that spans a broad semantic space (pairwise cosine similarities all below $<.75$; mean = 0.4, std = 0.086)
 - **Stage 2:** 250/250 sentences selected by the encoding model that were predicted to maximally drive/suppress the language network.
- Participants:
8 (Stage 1) + 5 (Stage 2)
50 sentences per run
24 runs in total (200 sentences repeated once)



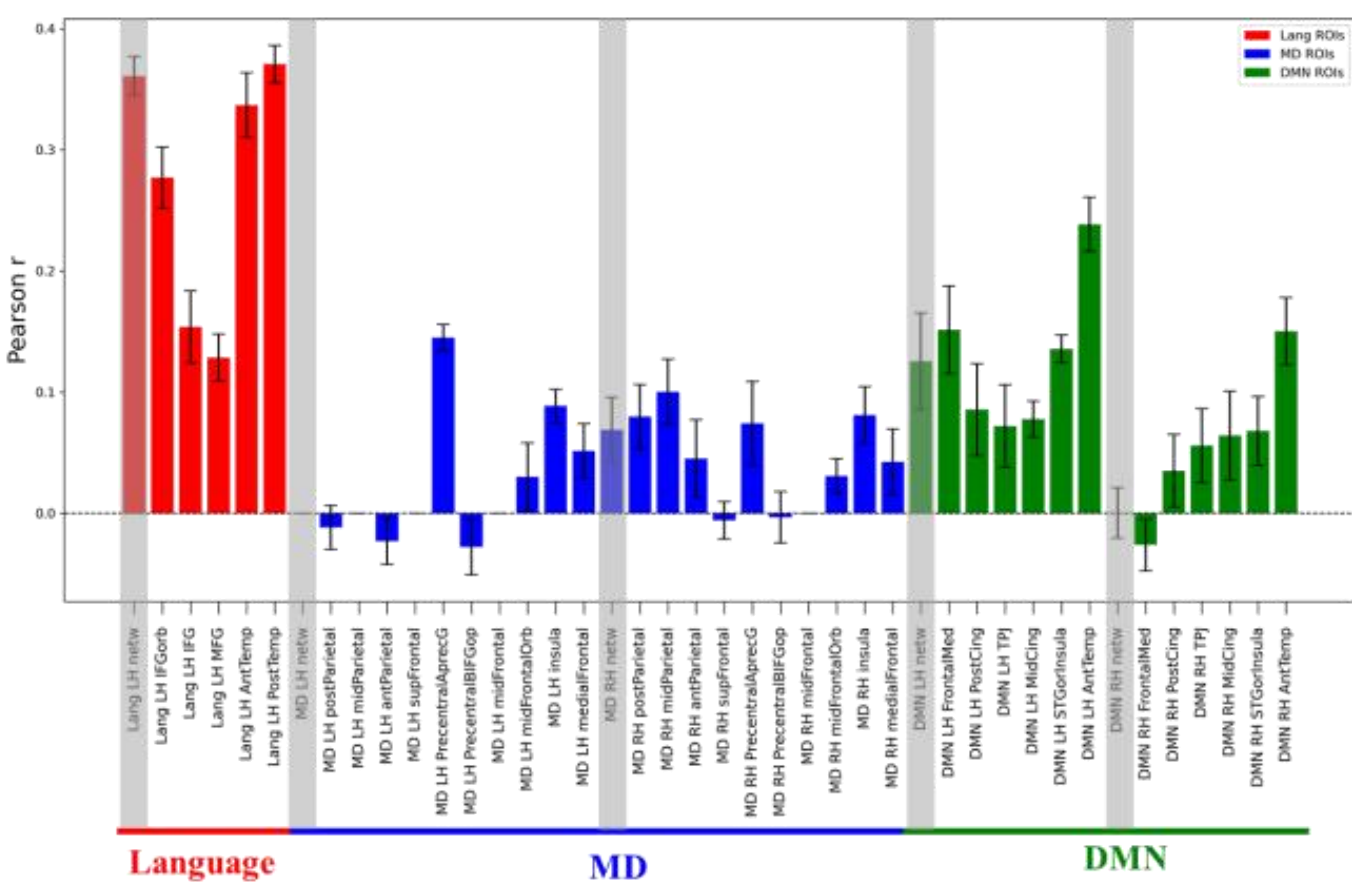
Methods



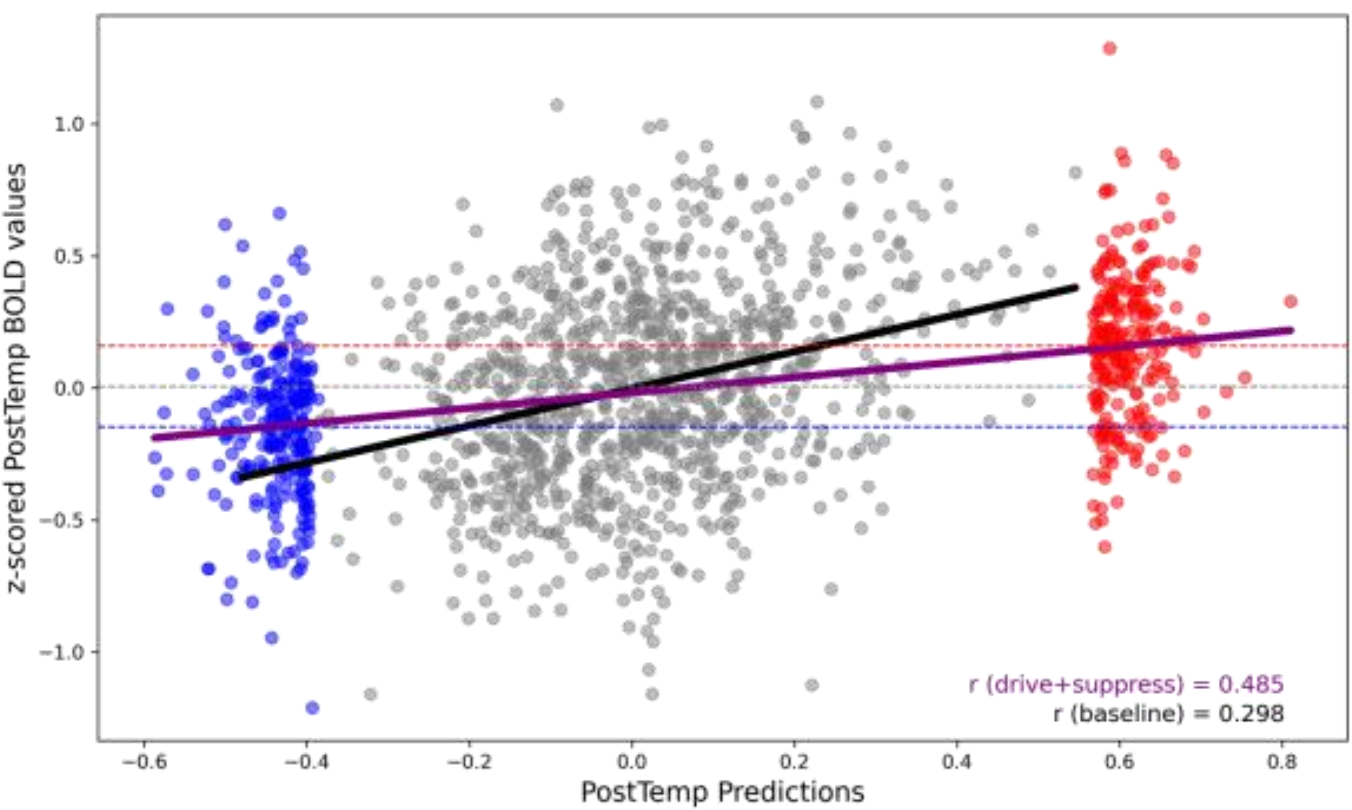
Constructed an encoding model by correlating the beta estimates from the participants' posterior temporal ROI with sentence word embeddings.

fMRI data from five additional participants were collected to evaluate predictive effectiveness of the model.

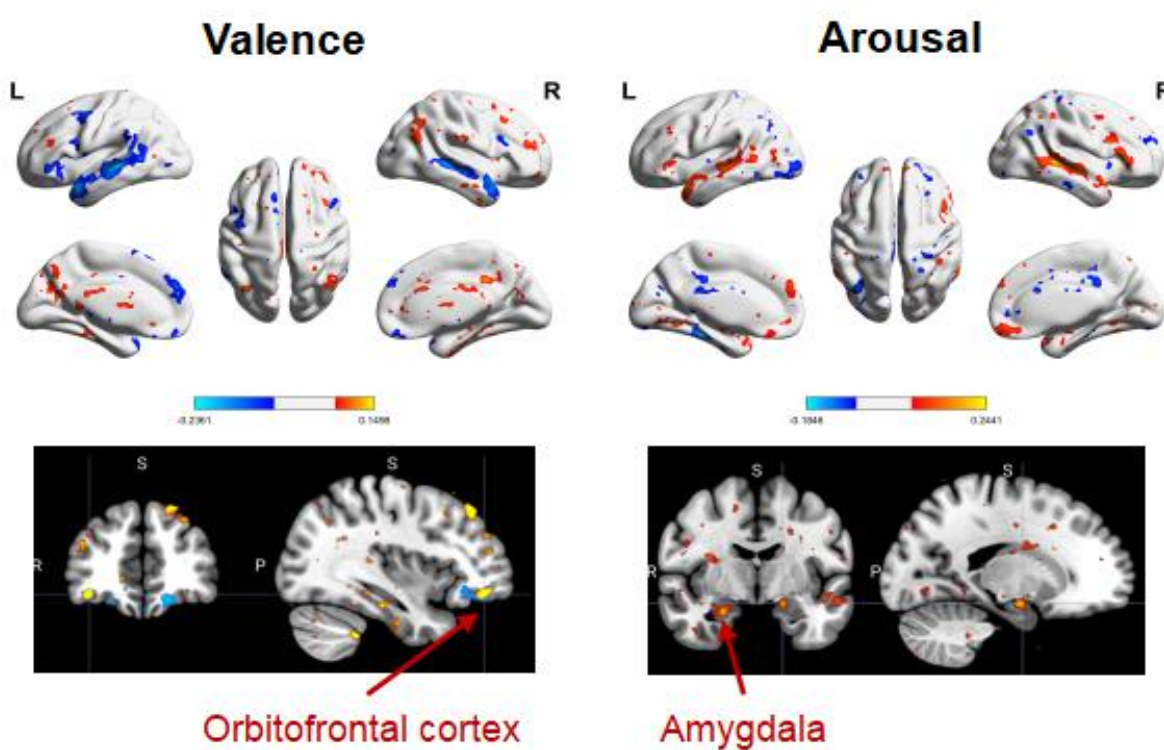
Results



5-fold CV, mean pearson r for language and control (MD,DMN) ROIs



Relationship between actual and predicted responses in a representative ROI (PostTemp)
=> model is good!



Widespread associations between valence or arousal and neural activity, suggesting distributed affective representations

Conclusions

- Brain responses to sentences **can be reliably predicted** using an encoding model grounded in sentence embeddings.
- **Emotional dimensions** such as valence and arousal systematically modulate neural activity within the language network, suggesting that the brain's representation of sentence meaning is **shaped by rich affective information** rather than isolated,category-specific mechanisms

References

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- 4.Mendes GA, Martins B. Quantifying Valence and Arousal in Text with Multilingual Pre-trained Transformers. arXiv. Preprint posted online February 27, 2023.

Contact

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